Computing the interleaving distance between two interval modules

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I found nowhere the explicit computation of the interleaving distance between interval modules, tough it is well known that it equals the bottleneck distance.

1 Preliminaries

Let $\mathbf{k}$ be an arbitrary field, fixed throughout. We denote by $\mathbb{R}$ either the real line or the category induced by the poset $(\mathbb{R}, \leq)$. All functors are covariant.

1.1 The category of persistence modules

A **persistence module** is a functor $\mathbb{R} \to \mathrm{Vect}(\mathbf{k})$.

Example 1 (Important)**.** Let $I \subseteq \mathbb{R}$ be an interval. Define $\mathbf{k}^I \colon \mathbb{R} \to \mathrm{Vect}(\mathbf{k})$ by

$$t \mapsto \begin{cases} \mathbf{k} & \text{if } t \in I \\ 0 & \text{else} \end{cases} \qquad \text{and} \qquad r \to s \ \mapsto \begin{cases} \mathbf{k} \xrightarrow{\mathrm{Id}} \mathbf{k} & \text{if } r, s \in I \\ 0 & \text{else} \end{cases}$$

on objects and morphisms, respectively. We call $\mathbf{k}^I$ the **interval module** over I.

Remark. Interval modules are the building blocks of persistence modules. In fact, one objective of persistence theory is to understand persistence modules by decomposing them into interval modules. However, this is not always possible. A situation when such a decomposition is guaranteed is when we work with finite dimentional vector spaces, that is, if we work over $\mathrm{vect}(\mathbf{k})$ instead of $\mathrm{Vect}(\mathbf{k})$.

The concise definition of persistence module can be expanded to aide comprehension. Let $V \colon \mathbb{R} \to \mathrm{Vect}(\mathbf{k})$ be a persistence module.

- For each $t \in \mathbb{R}$ and each arrow $r \to s$ denote $V(t) = V_t$ and $V(r \to s) = v_s^r$.
- Since V is a functor, we have the so-called **composition law**:

$$v_t^s \circ v_s^r = v_t^r \quad \text{whenever } r \leq s \leq t$$

  and $V(t \to t) = v_t^t = \mathrm{Id}_{V_t}$.

Thus, a persistence module V is a family of vector spaces $(V_t)_{t\in\mathbb{R}}$ together with a family $(v_t^s\colon V_s \to V_t \mid s \leq t)_{s,t\in\mathbb{R}}$ of $\mathbf{k}$-linear maps that satisfy the composition law.

We got a set theorical definition from the categorical one above. In fact, the two are equivalent. We will use the first, but prefer the notation of the second.

Since persistence modules are functors, we can consider the natural transformations between them, and since those can be composed (using vertical composition) we obtain a category.

Definition 1. The **category of persistence modules** is $\mathbf{X} = \mathrm{Fun}(\mathbb{R}, \mathrm{Vect}(\mathbf{k}))$.

Yes, this category is fun. Concretely, a morphism of persistence modules $\alpha\colon U \to V$ is a family of linear maps $(\alpha_r\colon U_r \to V_r)_{r\in\mathbb{R}}$ such that the diagram

$$\begin{tikzcd} U_s \arrow[r, "u_t^s"] \arrow[d, "\alpha_s"'] & U_t \arrow[d, "\alpha_t"] \\ V_s \arrow[r, "v_t^s"] & V_t \end{tikzcd}$$

commutes for all $s \leq t$.

1.2 The shift functor

Let U be a persistence module. Define a new persistence module $U[t]$ by

$$U[t](r) = U_{t+r} \quad \text{and} \quad U[t](r \to s) = u_{t+r}^{t+s}.$$

We say that $U[t]$ is a **shifted module**.

Suppose we've made two shifts of U, say $U[s]$ and $U[t]$, where $s \leq t$. We might ask ourserlves: what is the relationship between these shifted modules? Since these are functors, we wonder if there is a canonical natural transformation $\alpha\colon U[s] \to U[t]$, and in fact there is. To find out, consider any $p \to q$ and apply both $U[s]$ and $U[t]$ to this arrow. We get

$$U[s](p \to q) \;=\; U_{p+s} \xrightarrow{u_{q+s}^{p+s}} U_{q+s}$$

and

$$U[t](p \to q) \;=\; U_{p+t} \xrightarrow{u_{q+t}^{p+t}} U_{q+t}$$

Since $\alpha = (\alpha_r)_{r\in\mathbb{R}}$ is determined by its components, we need to find the vertical arrows that make the square

$$\begin{tikzcd} U_{p+s} \arrow[r, "u_{q+s}^{p+s}"] \arrow[d, "\alpha_p"'] & U_{q+s} \arrow[d, "\alpha_q"] \\ U_{p+t} \arrow[r, "u_{q+t}^{p+t}"'] & U_{q+t} \end{tikzcd}$$

commute. Set $\alpha_p = u_{p+t}^{p+s}$ and $\alpha_q = u_{q+t}^{q+s}$. Then the composition law give us

$$u_{q+t}^{p+t} \circ u_{p+t}^{p+s} = u_{q+t}^{p+s} = u_{q+t}^{q+s} \circ u_{q+s}^{p+s},$$

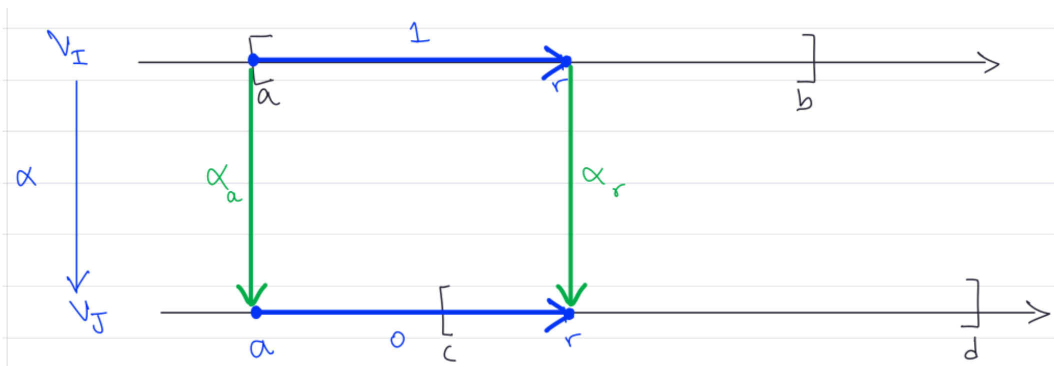
which means the square commutes. Thus α is indeed a natural transformation, that we'll denote $U[s \to t]$. To sum up, we have the following natural transformation

$$U[s \to t] = \left(u_{t+r}^{s+r}\right)_{r \in \mathbb{R}}.$$

On the other hand, note that

$$U[t][t'] = U[t + t']$$

for any $t, t' \in R$.



The functor $T = [\,\cdot\,] \colon \mathbb{R} \to \mathrm{End}(X)$ is defined as follows

- Objects: for each $t \in \mathbb{R}$ we have a functor $T(t) \colon X \to X$ denoted T_t, which is defined by $T_t(U) = U[t]$ on objects $U \in X$, and by $T_t(\alpha) = (\alpha_{r+t})_{r \in \mathbb{R}}$ on morphisms $\alpha = (\alpha_r)_{r \in \mathbb{R}}$. (Equivalently, we can define $T_t(\alpha) = \alpha \star [t]$)

- Morphisms: for each $r \to s$, define a natural transformation $T(r \to s) \colon T_r \to T_s$ by

$$T(r \to s) \colon T_r \to T_s = (U[r \to s])_{U \in X}$$

1.3 The interleaving distance between persistence modules

Let U and V be persistence modules. A t-**interleaving** of U and V is a pair of morphisms (ϕ, ψ) that make the diagrams

$$\begin{tikzcd} U \arrow[rr, "\mathbb{1}_U^{2t}"] \arrow[dr, "\phi"'] & & U[2t] \\ & V[t] \arrow[ur, "\psi[t]"'] & \end{tikzcd} \qquad \begin{tikzcd} V \arrow[rr, "\mathbb{1}_V^{2t}"] \arrow[dr, "\psi"'] & & V[2t] \\ & U[t] \arrow[ur, "\phi[t]"'] & \end{tikzcd}$$

commute. If such a pair exists, we say that U and V are t-interleaved.

Definition 2. The interleaving distance between two interval modules is

$$d_T(U,V) = \inf\left\{t \geq 0 \mid \text{ there exists a } t\text{-interleaving of } U \text{ and } V\right\}.$$

This is the smallest shift we'd need to do for...

2 The computation

We'll do the computation for the case of closed intervals. Let $I = [a,b]$ and $J = [c,d]$. Denote by $\mathcal{I} = \mathbf{k}^I$ and $\mathcal{J} = \mathbf{k}^J$ the corresponding interval modules.

Our strategy is to find some apropriate bounds for $d_T(U,V)$. Befor going into the search for the pair of morphisms that would make the triangles above commute, we need to know how are the morphisms between interval modules, because then we can characterize the morphisms between shifted interval modules.

2.1 Bound above

2.2 Bound below